

Pavan Kumar Etta

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of production experience designing distributed systems, cloud-native microservices, and real-time data pipelines at scale. Proven track record of measurably improving system performance reducing latency by **40%**, boosting ingestion throughput by **35%**, and accelerating query response times by **70%**. Pursuing M.S. in Computer Information Systems (**GPA 4.0**) at UMass Dartmouth. Passionate about building systems that handle millions of requests with high reliability and low latency.

SKILLS

Languages: Python, Java, C++, C#, Go, JavaScript, TypeScript

Backend & Distributed Systems: Microservices, Distributed Systems, REST APIs, Event-Driven Systems

Databases & Storage: MySQL, PostgreSQL, MongoDB, Microsoft SQL Server (MSSQL), DynamoDB, Redis

Cloud & Infrastructure: AWS (Lambda, EC2, API Gateway, S3, Redshift), Docker, Kubernetes, Microsoft Azure

Data Engineering & Distributed Processing: Apache Spark, PySpark, Hadoop, Stream Processing, Data Pipelines

Frameworks: Node.js, React, React Native, Angular, Next.js, Vue.js, Django, Flask, GraphQL, Redux, HTML, CSS/SCSS

Machine Learning & Data Science: NumPy, Pandas, Scikit-learn, OpenCV, PyTorch, TensorFlow, NLTK

DevOps & Tools: GitLab CI/CD, Kafka, Linux, Bash, HTTP, TCP/IP, JIRA, Jest, AJAX, jQuery

PROFESSIONAL EXPERIENCE

Software Engineer II | Rhenix Lifesciences LLP

Aug 2021 – Aug 2024

- Designed a microservices-based full-stack architecture using Python, Django, and AWS (Lambda, API Gateway, EC2, DynamoDB) to process large-scale wearable telemetry and enable scalable, production-ready systems.
- Shipped REST API-backed iOS and Android mobile apps (Google Play Store & Apple App Store) with full CI/CD, supporting **2,000+** active users with high availability SLAs.
- Improved data ingestion throughput by **~35%** by building high-throughput REST APIs and distributed pipelines integrating Fitbit, Apple Health, and Withings APIs, ensuring reliable and scalable data processing.
- Architected event-driven pipelines using AWS MSK (Kafka) to enable real-time, asynchronous data workflows, improving system scalability and fault tolerance through multi-partition streaming design.
- Reduced end-to-end system latency by **~40%** through fault-tolerant ingestion pipeline optimization adding scheduling, circuit breakers, and workflow parallelism to enable near-real-time analytics.
- Developed a wearable data analytics dashboard processing **50K+** daily health data points, reducing preprocessing time by **40%**, and improving data availability for downstream insights.

Graduate Research Assistant | University of Massachusetts, Dartmouth

Nov 2024 – Dec 2025

- Developed React and Angular web apps with Typescript to present medical research data. Implemented features for data querying and visualization using Plotly.js and backend powered by Django and Python.
- Reduced query response time by **70%** by redesigning MySQL schema with partitioning, compound indexes, and a Redis caching layer, significantly cutting database load under concurrent access.
- Containerized the full application stack with Docker, improving deployment consistency and contributing to a **25%** increase in active user adoption.

Lead Software Engineer | Cure Science Institute

Jun 2021 – Aug 2021

- Built real-time telemetry ingestion pipelines using Python and NumPy, processing sensor streams from embedded hardware system devices.
- Implemented signal preprocessing and feature extraction pipelines using Python and MATLAB, improving signal quality and system stability by **~30%**.
- Built an ML-based anomaly detection system using Scikit-learn achieving **~93%** accuracy, enabling automated equipment health monitoring and reducing manual inspection overhead.

Software Developer Intern | Rashtriya Ispat Nigam Limited

May 2019 – Jun 2019

- Analyzed large-scale industrial telemetry datasets using Python data processing pipelines, improving fault detection accuracy by **~20%**.
- Built preprocessing pipelines to clean and standardize high-volume sensor data for downstream analytics systems.

EDUCATION

University of Massachusetts, Dartmouth	North Dartmouth, MA
Master of Science - Computer Information Systems, Data Science, GPA: 4.0	Aug 2024 – May 2026
Indian Institute of Information Technology, Sricity	Chittoor, India
Bachelor of Technology - Electronics and Communication Engineering, GPA: 3.2	Aug 2017 – May 2021

PROJECTS

Water Quality Analysis	(NetCDF, Tableau, PyTorch, PySpark, Hadoop, GCP)
Processed 30GB+ environmental datasets using PySpark and Hadoop on GCP; trained LSTM models achieving an MAE of 1.45 for water distribution trend forecasting demonstrating big data processing at petabyte-relevant frameworks.	
Fault-tolerant sharded key-value store	(C++, Raft, SQLite)
Implemented a production-grade distributed KV store using the Raft consensus protocol, supporting sharded replication, leader election, log compaction, and fault-tolerant storage across distributed nodes directly mirrors distributed storage patterns used at FAANG scale.	
GPU Power Consumption Analysis	(PyTorch, TensorRT, nvidia-smi)
Developed algorithms to analyze GPU workload energy consumption using PyTorch and TensorRT, optimizing CPU/GPU configurations for improved efficiency in edge computing workloads.	